30 July 1974
SUPERSEDING
MIL-G-45951
6 June 1973

MILITARY SPECIFICATION SHEET

GUN, AUTOMATIC, 20MM: GAU-4/A

This specification is approval for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the gun described herein shall consist of this document and the issue in effect of specification MIL-G-45500.

1. SCOPE

1.1 This specification covers one type of multibarrel, gas operated, electrically fired, 20 millimeter (mm) automatic gun for installation in an aircraft armament pod system.

2. APPLICABLE DOCUMENTS

SPECIFICATIONS

Military

MIL-G-45500	-	Gun, Automatic, 20mm: General Specification For.

DRAWINGS

Us. Army Weapons Command

F11010287	-	Gun, Automatic, 20mm: GAU-4/A
IEL11010287	-	Index of Inspection Equipment Lists.

PUBLICATIONS

U.S. Army Weapons Command

P11010287	-	Packaging Data Sheet for Gun, Automatic; 20mm: GAU-4/A
IEL11010287	-	Index of Inspection Equipment Lists.

FSC 1005

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3. REQUIREMENTS

3.1 Material, construction and design. Guns and parts shall conform the material, construction and design requirements specified herein and on Drawing F11010287.

3.2 Clearing solenoid. The clearing solenoid shall be fastened securely to the housing and the electrical receptacle connector shall be securely fastened to the connector bracket as specified on the applicable drawing.

3.3 Gas drive group. The spline shaft shall be securely retained in the end plate by the bolt. The cam shaft shall be securely attached to the ball spline by the retaining ring and shall be securely retained in place by the lock plate. The matched rear cap and the forward cap shall be securely attached to the gas cylinder. The gas cylinder shall be securely attached to the barrels by the cylinder clamp assemblies and bolts. There shall be no relative movement of the gas chamber or the clamps on the barrels. The inner and outer clamps shall be securely attached to the number 1 and 6 barrels by the bolts. There shall be no relative movement of the clamps on the barrels and the cam roller shall rotate on the inner clamp without binding. The ball spline, cam shaft and cam roller shall move through their full range of travel without binding when the barrel cluster is manually rotated.

3.4 Diffuser assembly. The diffuser assembly shall be fastened securely to the barrels as specified on the applicable drawing. There shall be no relative movement of the diffuser assembly or the barrels which are engaged by the muzzle clamp nut assembly.

3.5 Functioning.

TABLE I

Performance Requirements

Gas drive output torque (min)	Acceleration time (max.)	Steady state rate of fire	Stopping time (max.)
1800 pound-inch or brake torque (Consumed minimum 100 pounds-inch at minimum steady state rate of fire).	0.4 seconds (from start of gun rotation to minimum steady state rate of fire).	6,000 to 6,360 rounds per minute.	0.5 seconds (from minimum steady state rate of fire to complete stop).

3.5.1 Gun operating torque. The gun operating torque shall be a maximum of 800 pound-inches at minimum steady state rate of fire.

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3.6 Targeting and accuracy. At a range of 1,000 inches, 80 percent of a 100 round burst fired from the gun shall be completely within an 8.0 inch diameter circle for accuracy; and the center of the 80 percent group shall be within 2.5 inches elevation or depression and 4.75 inches azimuth right or left of the true boresight point for targeting.

3.7 Endurance. Guns shall fire 30,000 rounds for endurance. A series is 1,000 rounds using five (5) bursts of 200 rounds each.

TABLE II

Schedule of Replacement parts

Part nomenclature	Part number	Gun rounds + 1,000 rounds
Pin, spring (for block, locking	11698266	15,000
Shaft assembly, bolt	7790708	15,000
Follower, cam	11699894	15,000
Insulator assembly, firing pin	11010159	15,000
Insulator, pin, cam, firing	11010160	15,000
Pin, firing	7268303	15,000
Stop, contact assembly	7268730	15,000
Pin, cam, firing	7268304	15,000
Spring, firing pin	7268757	15,000

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality conformance inspection.

4.1.l Examination. Guns shall be visually and manually examined in accordance with the following:

TABLE III

Parts or assembly	Reference paragraph
Clearing solenoid	3.2
Gas drive group	3.3
Diffuser assembly	3.4

4.1.2 Inspection equipment. The measuring and testing equipment shall be as specified on the Index of Inspection Equipment List IEL11010287.

4.1.2.1 Mount. The function firing and endurance firing tests shall be accomplished with the guns mounted in a SUU-23/A armament pod or equipment of the contractor's design and approved by the Government simulating the function of the pod.

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4.1.2.2 Electrical power supply. The electrical power supply and control equipment shall be capable of supplying, to the pod input or test equipment 208 volts alternating current (line to line), 3 phase, 4 wire, 400 cycles per second frequency with a peak transient current of 35 amperes average for 0.1 seconds per phase (during the inertia motor starting) and a steady state current of 10 amps per phase (during flywheel acceleration). The voltage and frequency shall be maintained within plus or minus 5%. The inertia starter with the help of the gun gas drive shall meet the requirement for acceleration specified in Table I. The governor and its associated electronic control shall provide control to maintain steady state rate of fire and stopping time within the limits specified herein in Table I. A frequency discriminator from the governor control electronics shall be used to operate the gun automatic clearing solenoid at a gun rate of 4,100 plus or minus 50 rounds per minute.

4.2 Test methods.

4.2.1 Functioning test. Testing of guns for functioning, clearing, and gun operating torque requirements shall be accomplished in accordance with MIL-G-45500.

4.2.1.1 Gun operating torque test. Testing of guns for operating torque requirement shall be as specified herein. The gun gas drive unit shall be removed from the gun and slave barrels conforming to Drawings D7791616 and C11010336 shall be used for testing gun operating torque (see 6.1). Drives used for measuring gun operating torque shall be re-calibrated after approximately each 15,000 rounds. The 100-round burst plus or minus 10 rounds shall be fired to test for gun operating torque specified herein, and targeting and accuracy (see MIL-G-45500). The gun shall be functioned at a steady state rate of 6,000 to 6,360 rounds per minute using the power supply with control equipment, feeder, and calibrated drive.

4.2.1.2 Function firing test procedure. After the clearing test, gun operating torque test, and high-pressure resistance test are completed, the gun shall be mounted in accordance with 4.1.2.1 using electrical power supply in accordance with 4.1.2.2 specified herein.

4.2.1.2.1 The armament pod shall be loaded with a full complement of ammunition before testing each gun. No bursts shall be fired with less than 700 rounds remaining in the system. A 100-round burst plus or minus 10 rounds shall be fired from the gun to test for acceleration and stopping time, steady state rate of fire, brake torque or gas drive output, automatic clearing action, and functioning. At the completeion of firing, the gun shall be checked to assure that all the breech bolt assemblies are in the cleared position.

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4.2.1.2.2 Instrumentation shall be such that gun speed curve, timing marks, primer impulse, and brake torque or gas drive output torque. is taken and recorded concurrently for use in computing required data.

4.2.2 Endurance test. Testing of guns for endurance requirement (see 3.4.6) shall be accomplished with a gun held in a mount in accordance with 4.1.2.1 using electrical power supply in accordance with 4.1,2.2, specified herein.

4.2.3 Interchangeability test. The following groups of nonmating parts shall be utilized in the interchangeability test.

GROUPS OF NONMATING PARTS

GROUP I

Adapter, recoil, assembly (7268312)
*Barrels (11700001 through 11700006)
Block, locking (7269955)
Follower, cam (11699894)
Gear, rotor (7268664)
Housing assembly, rear (11010476)
Insulator, in, cam, firing (11010060)
Pin firing (7268303)
Ring, retainer (11699891)
Shaft, spline (11699883)
Stop, contact assembly (7268730)
Track, rotor center (726870)
Track, rotor front (7268711)
Track, rotor removable (7268552)

GROUP II

Bar, guide (11010130)
Cam, unlocking, front (11698265)
Clamp assembly (11699899)
Clamp, outer (11699898)
Cylinder, gas (11699886)
Insulator assembly, firing pin (11010159)
Pin, cam, firing (7268304
Pin, cover lock (11010132)
Plate, end (11699888)
Retainer assembly (7269886)
Shaft assembly, bolt (7790708)
Spacer, removable track (7268715)
Spline, ball (11699884)
Spring, firing pin (11698237)
Spring, helical, torsion (7791606)

GROUP III

Body, breech bolt (1101058)
Cam, lock (7790681)
Cam, unlocking, rear (11698264)
Clamp, coupling (7269760)
Clamp, inner (11700007)
Contact assembly, firing (11010186)
Cover, housing assembly (7268556)
Diffuser assembly (11698231)
Pin assembly, indexing (7790768)
Rotor assembly, (remaining parts)
Sector, assembly, automatic clearing (11010219)
Shaft, cam (11699882)

*Barrels shall be assembled in their respective positions.

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5. PREPRATION FOR DELIVERY

Packaging Data Sheet P11010287 applies.

6. NOTES

6.1 Ordering data. Procurement document should specify the following:

a. Responsibility for furnishing armament pods (see 4.1.2.1).

Custodians:
Army - WC
Navy - OS
Air Force - 84

Review activities:
Navy - OS
Air Force - 84

User activity:
Navy - MC

Preparing activity:
Army - WC

Project number:
1005-0525-2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-45500/2: Gun, Automatic, 20MM: GAU-4/A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRA

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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MIL-G-45500/2
AMENDMENT
3 OCTOBER 1988

SUPERSEDING
MIL-G-45500/2
AMENDMENT 1
5 OCTOBER 1977

MILITARY SPECIFICATION SHEET

GUN, AUTOMATIC, 20MM: XM130

This amendment forms a part of Military specification Sheet MIL-G-45500/2, dated 30 July 1974, and is approved for use by all Departments and Agencies of the Department of Defense.

Throughout the specification, delete "GAU 4/A" and substitute "XM130"

PAGE 3

Delete "TABLE II" in its entirety.

Add:

"TABLE II"

"SCHEDULE FOR REPLACEMENT PARTS (NOT APPLICABLE)".

PAGE 4

Para 4.2.1.1, Line 3: Delete "conforming to drawings D7791616 and C11010336" and substitute ", mid-barrel clamp and muzzle clamp, conforming to drawings F7790801, D7791616 and C11010366 respectively,".

PAGE 5

Para 4.2.3, Group I -

Delete "7269955" and substitute "11691480"

Delete "7268303" and substitute "11691418"

Group II

Delete "7268304" and substitute '11691417"

Delete "7790707" and substitute "11691421"

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Group III

Delete “1101058” and substitute “11691423”

Para 4.2.3, Group I -

Delete “11010476” and substitute “12007363”

Delete “Insulator, Pin, Cam, Firing (11010060) and substitute “Insulator, Vertical (11691415)”

Delete “Stop, Contact Assembly (7268730)” and substitute “Plug (11691414)”

Group II

Delete “Insulator Assembly, Firing Pin (11010159)” and substitute “Insulator Assembly, Horizontal (11691420)”

Delete “Pin, Cam, Firing” and substitute “Pin, Cam”

Delete “Spring, Firing Pin (11698237) and substitute “Spring, Compression, Helical (11691419)”

Group III

Delete “7790681” and substitute “11691677”

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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